A project report on

EXPLAINABLE AI FOR EARLY FETAL DISTRESS DETECTION: A SHAP-ENHANCED CTG ANALYSIS

Submitted in partial fulfillment for the award of the degree of

M.Tech. (Integrated) in Software Engineering

by

MINNA FATHIMA (21MIS1070)

VIT®
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

April, 2026

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DECLARATION

I here by declare that the thesis entitled “EXPLAINABLE AI FOR EARLY FETAL DISTRESS DETECTION: A SHAP-ENHANCED CTG ANALYSIS” submitted by me, for the award of the degree of M.Tech. (Integrated) in Software Engineering, Vellore Institute of Technology, Chennai, is a record of bonafide work carried out by me under the supervision of “Dr. PREM SANKAR N”.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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CERTIFICATE

This is to certify that the report entitled **"Explainable AI For Early Fetal Distress Detection: A Shap-Enhanced CTG Analysis"** is prepared and submitted by **Minna Fathima** (**21MIS1070**) to Vellore Institute of Technology, Chennai, in partial fulfillment of the requirement for the award of the degree of **M.Tech. (Integrated) in Software Engineering** is a bonafide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

Signature of the Guide:

Name: Dr./Prof. Date:

Signature of the Examiner 1 Signature of the Examiner 2

Name: Name:

Date: Date:

Approved by the Head of Department

ABSTRACT

Accurate assessment of fetal health is essential for reducing preventable risks associated with pregnancy and childbirth. This study addresses the problem of automated fetal health classification using Cardiotocography (CTG) data obtained from Kaggle. A systematic machine learning approach is employed, involving data preprocessing, feature scaling, and handling class imbalance using SMOTE, followed by classification using a Random Forest model. To enhance transparency and reliability in clinical decision-making, SHAP-based explainable AI techniques are integrated to interpret model predictions. The proposed approach achieves an overall accuracy of approximately 91%, with strong ROC-AUC scores across all classes, demonstrating effective discrimination between normal, suspect, and pathological conditions. The results indicate high reliability in identifying critical pathological cases, which is crucial for early intervention. The study concludes that combining robust classification techniques with interpretability methods can provide a dependable and transparent framework for fetal health assessment. These findings highlight the potential of such systems to support healthcare professionals in making informed decisions and improving prenatal care outcomes, particularly in resource-constrained environments.

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Place: Chennai

Date: **Name of the Student**

CONTENTS

CONTENTS.....iii

LIST OF FIGURES.....v

LIST OF TABLES.....vi

LIST OF ACRONYMS.....vii

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION.....1

1.2 OVERVIEW.....2

1.3 CHALLENGES FACED.....3

1.4 PROBLEM STATEMENT.....4

1.5 OBJECTIVES.....5

1.6 SCOPE OF THE PROJECT.....5

CHAPTER 2

BACKGROUND

2.1 INTRODUCTION.....6

2.2 MOTIVATION.....6

2.3 TECHNICAL EVOLUTION.....7

2.4 LITERATURE SURVEY.....9

CHAPTER 3

ARCHITECTUREDIAGRAM

3.1 ARCHITECTURE DIAGRAM.....17

CHAPTER 4

CODE SNIPPETS

4.1 CODE SNIPPETS 23

CHAPTER 5

RESULTS AND DISCUSSION

5.1 TRAINING PHASE RESULTS 27

5.2 VALIDATION AND INFERENCE PERFORMANCE 27

5.3 SHAP VISUALIZATION FINDINGS 31

5.4 QUANTITATIVE REPORT INTERPRETATION 37

5.5 RECOMMENDED CLINICAL ACTIONS 37

5.6 SYSTEM EVALUATION 37

5.7 ISSUE ANALYSIS AND SOLUTIONS 39

5.8 DISCUSSION SUMMARY 40

5.9 FINAL OUTCOME 40

CHAPTER 6

PRACTICAL AND CLINICAL IMPLICATIONS

6.1 CLINICAL RELEVANCE 41

6.2 INTEGRATION WITH HIS 41

6.3 POTENTIAL MARKET…..... 42

6.4 VISUALIZATION OF MODEL FAILURE CASES 43

6.5 REAL-WORLD CASE STUDIES 44

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION 45

7.2 FUTURE WORK 46

LIST OF FIGURES

1.1 WORKFLOW DIAGRAM 2

3.1 SYSTEM ARCHITECTURE 16

5.1 CONFUSION MATRIX… 28

5.2 CLASSIFICATION REPORT 28

5.3 ROC CURVE 29

5.4 PRECISION – RECALL CURVE 30

5.5 FEATURE IMPORTANCE 30

5.6 GLOBAL SHAP VISUALIZATIONS 31

5.7 SHAP BEESWARM ANALYSIS 32

5.8 STREAMLIT DASHBOARD 33

LIST OF TABLES

2.1 LITERATURE SURVEY 11

5.1 PERFORMANCE BENCHMARKS 38

5.2 ISSUE ANALYSIS & SOLUTIONS 39

6.1 ERROR TYPE,OBSERVATION, & CAUSE 43

LIST OF ACRONYMS

Acronym	Full Form
CTG	Cardiotocography
AI	Artificial Intelligence
ML	Machine Learning
XAI	Explainable Artificial Intelligence
SHAP	SHapley Additive exPlanations
FHR	Fetal Heart Rate
STV	Short-Term Variability
RF	Random Forest
SMOTE	Synthetic Minority Oversampling Technique
CSV	Comma-Separated Values
PDF	Portable Document Format
UI	User Interface
ROC	Receiver Operating Characteristic
AUC	Area Under Curve
NICE	National Institute for Health and Care Excellence
IPOC	Integrated Plan of Care
bpm	Beats Per Minute
XGBoost	Extreme Gradient Boosting
API	Application Programming Interface
GPU	Graphics Processing Unit

Chapter 1

Introduction

1.1 INTRODUCTION

Fetal health assessment is a fundamental component of prenatal care, as it plays a vital role in identifying potential risks that may affect the well-being of both the mother and the fetus. Complications arising during pregnancy, if not detected at an early stage, can lead to severe outcomes including fetal distress, preterm birth, or even mortality. Therefore, continuous monitoring and timely diagnosis are essential to ensure safe pregnancy outcomes. One of the most widely used techniques for monitoring fetal condition is Cardiotocography (CTG), which records fetal heart rate (FHR) and uterine contractions over a period of time. These physiological signals provide critical information about the oxygen supply and neurological status of the fetus.

Despite being widely used, CTG data interpretation is frequently difficult and necessitates a high level of clinical competence. Manual analysis is subjective and time-consuming because the patterns seen in CTG signals can vary greatly based on a number of factors. The same CTG trace may be interpreted differently by different clinicians, which could result in inconsistent diagnoses. Because of this subjectivity, automated and standardized methods are required to help healthcare workers make consistent and accurate decisions.

With the rapid advancement of machine learning and artificial intelligence, there has been increasing interest in applying these techniques to healthcare problems. Machine learning models have the capability to learn complex patterns from data and make predictions with high accuracy. In the context of fetal health monitoring, these models can analyze CTG-derived features and classify fetal conditions into different categories. Furthermore, the use of explainable AI approaches makes it possible to comprehend prediction processes better, which is critical in medical applications where trust and transparency are vital.

1.2 OVERVIEW

This project follows a structured pipeline for fetal health classification using CTG data obtained from Kaggle. The architecture is designed to ensure accurate prediction, data handling, and interpretability.

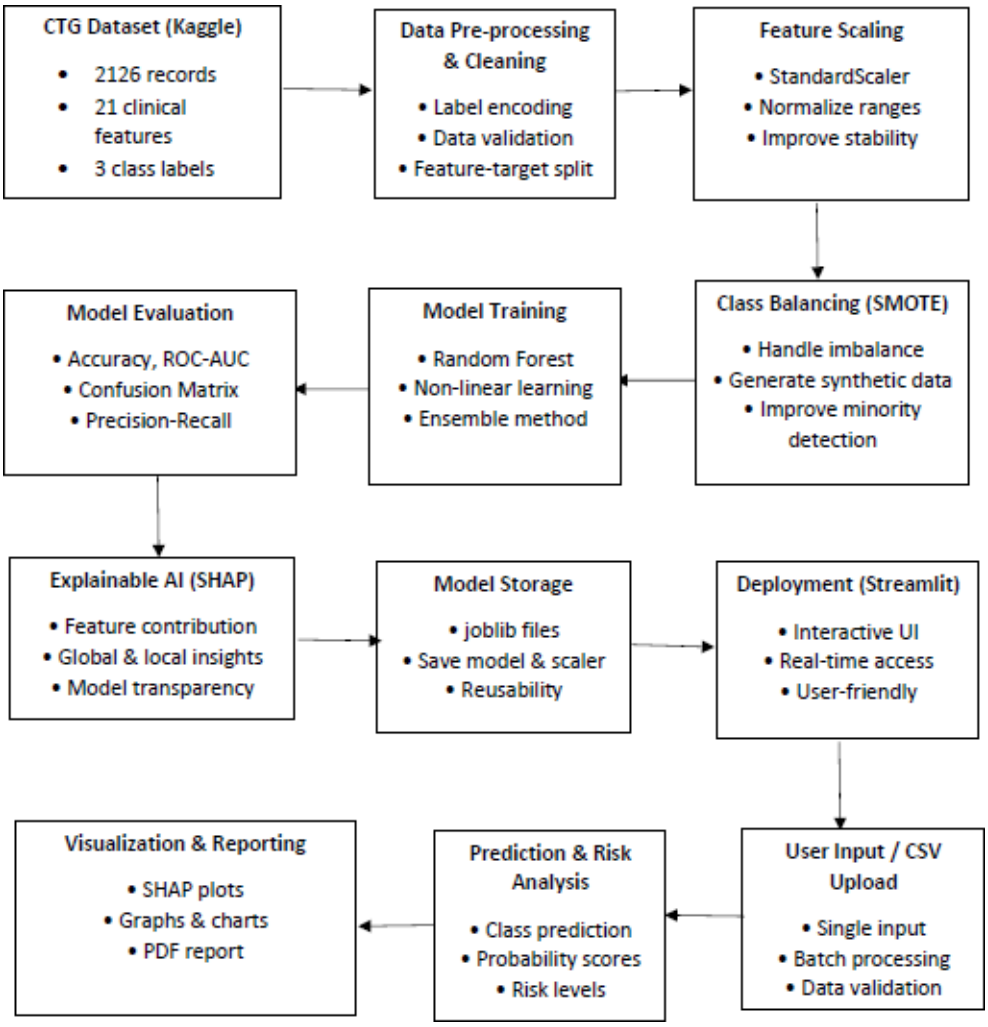

Fig 1.1 Workflow diagram

Explanation:

1. The system begins with a CTG dataset containing clinically derived features representing fetal heart rate patterns and uterine activity.
2. Data preprocessing is performed to clean the dataset, encode labels, and separate input features from the target variable.
3. Feature scaling is applied to normalize the range of values, ensuring stable and efficient model learning.
4. To address class imbalance, SMOTE is used to generate synthetic samples for minority classes, improving model fairness.
5. A Random Forest classifier is trained to learn complex patterns and perform multi-class classification of fetal health conditions.
6. The trained model is evaluated using performance metrics such as accuracy, ROC-AUC, and confusion matrix to ensure reliability.
7. An Explainable AI layer (SHAP) is integrated to analyze feature contributions and provide transparency in predictions.
8. The model and preprocessing components are stored for reuse, enabling efficient deployment without retraining.
9. The system is deployed through a Streamlit interface, allowing users to interact with the model easily.
10. Users can provide single inputs or upload datasets, and the system generates predictions along with probability scores and risk levels.
11. Final outputs include visualizations, SHAP-based explanations, and automated reports, making the system suitable for practical decision support.

1.3 CHALLENGES FACED

The development of this system involved addressing several technical and practical challenges to ensure accuracy, reliability, and usability.

- **High-dimensional clinical data**: The management of multiple CTG features necessitated careful comprehension and selection to prevent noise from compromising model performance.
- **Class imbalance in dataset**: Unequal distribution of Normal, Suspect, and Pathological classes posed a challenge in achieving a fair prediction across all categories.
- **Feature variability and scaling issues**: Differences in value ranges among features required proper normalization to ensure stable model training.

- **Model selection and optimization**: Identifying a suitable algorithm that balances accuracy, interpretability, and computational efficiency required experimentation.
- **Ensuring model generalization**: Maintaining high accuracy on unseen data while preventing overfitting was a critical challenge.
- **Interpretability in medical context**: Providing meaningful explanations for predictions was necessary to ensure trust and usability in healthcare scenarios.
- **Integration of multiple components**: Combining preprocessing, modeling, explainability, and deployment into a single pipeline required careful coordination.
- **Real-time usability considerations**: Designing a system that supports both single input and batch processing while maintaining responsiveness.
- **Visualization and result presentation**: Presenting outputs in a clear, understandable format for users without technical background required thoughtful design.
- **Deployment and reproducibility**: Ensuring consistent results across environments and enabling smooth deployment posed practical challenges.

1.4 PROBLEM STATEMENT

The assessment of fetal health using Cardiotocography (CTG) data involves complex interpretation of multiple physiological parameters, which can be time-consuming and dependent on expert judgment. Variability in analysis and the need for continuous monitoring increase the risk of delayed or inconsistent decision-making. There is a need for a reliable computational approach that can process multidimensional clinical data efficiently and support accurate classification of fetal conditions. The difficulty is in the creation of a system that not only achieves high predictive performance but also guarantees the interpretability and consistency of the results. Consequently, the objective of this research is to develop a data-driven model that can aid in the early identification of potential risks, thereby facilitating the development of informed and expeditious clinical decisions.

1.5 OBJECTIVES

- To develop a machine learning model that can accurately classify fetal health conditions by utilizing clinical features derived from CTG.
- To integrate Explainable AI using SHAP (SHapley Additive exPlanations) to provide a transparent, feature-level interpretation of model predictions.
- To gain a more comprehensive understanding of the factors that influence fetal health outcomes by analyzing the importance of both global and local features.
- To develop a system that generates interpretable and clinically meaningful outputs, thereby facilitating real-world medical decision-making.
- To ensure the model delivers consistent and reliable predictions suitable for practical deployment scenarios.
- To build a solution that is capable of real-time usage and batch processing, enhancing its usability in diverse healthcare environments.
- To generate outputs that include risk-level insights and visual explanations, making the system accessible to non-technical users.
- To create a scalable framework that can be extended to real-world clinical applications and decision-support systems.

1.6 SCOPE OF THE PROJECT

The scope of this project focuses on building a data-driven system that utilizes CTG-derived features to perform structured analysis and classification of fetal health conditions within a controlled computational environment. It underscores the importance of combining machine learning with explainable AI to offer both predictive outcomes and interpretable insights, thereby facilitating a meaningful comprehension of the influence of features. Flexible input management, including individual and batch-level data processing, as well as visual outputs for improved interpretability, are part of the system's design. The framework is adaptable and can be extended to analogous clinical datasets or decision-support applications, despite the fact that the implementation is based on a validated dataset from Kaggle. The initiative establishes a foundation for future real-world integration, but it remains within the scope of analytical modeling and system development, without direct clinical deployment.

Chapter 2

Background

2.1 INTRODUCTION

Fetal health monitoring is a critical component of prenatal care, aimed at ensuring the well-being of both the mother and the developing fetus. One of the most widely used diagnostic tools is Cardiotocography (CTG), which records fetal heart rate and uterine contractions to assess fetal condition. Nevertheless, the interpretation of CTG signals necessitates clinical expertise and can be subjective, resulting in potential variability in the diagnosis. Assisted by the development of data-driven approaches and machine learning, automated systems can facilitate the analysis of intricate physiological patterns, thereby facilitating the quicker and more consistent classification of fetal health into categories such as normal, suspect, or pathological.

2.2 MOTIVATION

The primary motivation for this project is to enhance the reliability of fetal health assessment and decrease the reliance on subjective interpretation through the use of intelligent systems. In numerous healthcare environments, particularly those with restricted access to experienced specialists, delayed or inaccurate diagnoses can result in severe complications. This project endeavors to offer transparent reasoning behind each decision in addition to accurate predictions by combining machine learning with explainable AI techniques like SHAP. This ultimately contributes to enhanced maternal and fetal outcomes by supporting better-informed medical decisions and building trust among clinicians.

2.3 TECHNICAL EVOLUTION

Technical advancements in fetal health risk monitoring and assessment

Fetal health monitoring has undergone rapid transformation with the integration of artificial intelligence, advanced sensing technologies, and modern diagnostic approaches. The emphasis has shifted from subjective and intermittent monitoring toward continuous, objective, and predictive systems that facilitate early detection of fetal distress and improve clinical decision- making.

1. Artificial Intelligence and Machine Learning Integration

• The capacity to analyze complex, high-dimensional CTG data has been substantially enhanced by Artificial Intelligence (AI), thereby reducing the need for manual interpretation.
• Advanced algorithms are employed by computerized CTG (cCTG) systems to assess both conventional features (e.g., baseline heart rate and variability) and advanced signal-based parameters.
• Deep learning architectures, including Convolutional Neural Networks (CNNs) and Gated Recurrent Units (GRUs), facilitate the direct analysis of raw neonatal heart rate signals, thereby capturing subtle temporal patterns.
• Real-time risk prediction is facilitated by AI-based systems, which facilitate proactive surveillance rather than reactive intervention.
• The integration of Explainable AI techniques improves transparency, enabling clinicians to interpret model decisions with greater confidence.

2. Advanced Non-Invasive Monitoring Technologies

• The objective of emerging technologies is to offer fetal surveillance that is both continuous and comfortable, in addition to conventional Doppler ultrasound techniques.
• The reliability of signals is enhanced by the use of abdominal sensors in fetal electrocardiography (fECG), which allows for the non-invasive measurement of fetal cardiac activity.
• Direct indicators of fetal oxygenation and distress are provided by innovations in biochemical sensing, such as lactate monitoring.
• Home-based monitoring is facilitated by wearable IoT-based devices, which can continuously monitor embryonic movements through the use of embedded motion sensors.
• These developments improve accessibility and decrease dependence on hospital-based equipment.

3. Advancements in Non-Invasive Prenatal Testing (NIPT)

- Cell-free fetal DNA (cfDNA) from maternal blood is employed in contemporary NIPT techniques to facilitate the early detection of genetic abnormalities.
- Improved accuracy is achieved through the use of expanded testing methodologies to identify microdeletions, duplications, and chromosomal variations.
- Diagnostic capabilities are expanded beyond conventional screening by emerging approaches, which facilitate the identification of single-gene disorders.
- These technologies aid in the early identification of risks and the planning of preventive care.

4. Telemedicine and Remote Monitoring Systems

- Telemedicine has facilitated remote fetal monitoring, particularly for high- risk pregnancies.
- Continuous monitoring outside of clinical contexts is facilitated by mobile CTG systems, which enhances accessibility in low-resource environments.
- Clinicians can remotely monitor patients with the assistance of cloud-based platforms, which facilitate real-time data transmission and analysis.
- Remote telemetry systems facilitate timely intervention and minimize superfluous hospital visits.
- These systems are particularly beneficial in enhancing maternal and fetal outcomes in underserved regions.

Future Directions and Emerging Technologies

- The objective of research into artificial womb systems is to provide support for exceedingly premature infants, which is a significant advancement in neonatal care.
- Novel insights into metabolic activity and embryonic brain development are facilitated by advancements in functional imaging and fetal MRI.
- It is anticipated that the predictive accuracy will be improved through the integration of multimodal data sources (CTG, imaging, genomics).
- The advancement of automated decision-support systems will provide clinicians with additional support in intricate situations.
- Real-time intelligence, scalability, and clinical integration will be prioritized in future systems.

2.4 LITERATURE SURVEY ON PROPOSED PROJECT

2.4.1. Overview

Fetal health assessment has gained significant attention in recent years due to its critical role in reducing maternal and neonatal complications. Cardiotocography (CTG) is widely used to monitor fetal heart rate and uterine contractions, enabling clinicians to classify fetal conditions into normal, suspect, and pathological categories. However, traditional interpretation of CTG data is often subjective and time-consuming. Recent studies have demonstrated the effectiveness of machine learning and deep learning techniques in automating fetal health classification with high accuracy. Models such as Random Forest, Support Vector Machine, XGBoost, and deep neural networks have shown promising results in identifying fetal distress and improving diagnostic efficiency.

2.4.2. System Concept

Intelligent decision-support tools that incorporate data preprocessing, machine learning models, and classification mechanisms are the design of contemporary fetal health prediction systems. These systems typically employ CTG datasets that comprise a variety of physiological features, including fetal heart rate variability, accelerations, and decelerations. SMOTE is frequently implemented to mitigate class imbalances, while ensemble models like Random Forest and LightGBM are implemented for classification purposes due to their exceptional performance and durability. The objective of these systems is to facilitate the timely identification of abnormalities, thereby enhancing pregnancy outcomes and facilitating clinical intervention.

2.4.3. Integrated Interpretability Layer

In healthcare, the "black-box" nature of conventional machine learning models is one of the primary limitations, as it undermines clinicians' confidence. In order to resolve this matter, explainable artificial intelligence (XAI) methodologies, including SHAP (SHapley Additive Explanations) and LIME, have been extensively implemented. SHAP, in particular, has become the predominant method for interpreting model predictions by quantifying the contribution of each feature. Research indicates that the integration of interpretability in healthcare systems enhances the reliability of decision- making, supports clinical validation, and improves transparency. This layer facilitates the provision of both global and local explanations, thereby assisting clinicians in comprehending the reasoning behind the prediction outcome.

2.4.4. Automated Report Generation

The automated report generation module transforms model outputs into structured and clinically meaningful reports.

- Converts model predictions into structured, human-readable clinical reports.
- Includes patient details, CTG parameters, predicted diagnosis, and probability distribution.
- Provides clear clinical interpretation based on prediction results.
- Reduces manual effort and ensures consistency in medical reporting.
- Improves communication between AI system and healthcare professionals.
- Supports quick decision-making in critical situations.

2.4.5. Implementation pipeline

The implementation pipeline outlines the step-by-step process followed to develop and deploy the fetal health prediction system.

- Data collection from CTG datasets containing fetal health parameters.
- Data preprocessing including cleaning, normalization, and feature selection.
- Handling class imbalance using techniques such as SMOTE.
- Splitting data into training and testing sets for evaluation.
- Training machine learning models (e.g., Random Forest).
- Evaluating model using metrics like accuracy, precision, recall, and ROC- AUC.
- Integrating explainability techniques (SHAP) for interpretation.
- Deploying the model using a web-based interface (Streamlit).

2.4.6. Novelty and Strengths

The proposed system offers several unique features and advantages compared to traditional fetal health prediction approaches.

- Combines machine learning with explainable AI (SHAP) for transparency.
- Provides individual prediction and dataset-level explanations.
- Includes batch processing for multi-patient analysis.
- Integrates interactive visualizations for better understanding.
- Generates automated clinical reports in PDF format.
- Enhances reliability and trust in AI-based medical systems.
- Designed as a complete end-to-end decision support system.

2.4.7 Literature Survey on Recent Advancements in Fetal Distress Detection Technologies

Table 2.1 Literature Survey

Sl. No.	Author & Year	Method / Approach	Key Contribution	Gap Identified
1	Kaya T., Kaya D., Atar F. (2025)	Deep learning–based feature extraction using AlexNet, combined with metaheuristic optimization algorithms (Whale Optimization Algorithm and War Strategy Optimization) for feature selection and SVM classification.	Proposed a hybrid model that converts CTG signals into 2D representations and selects optimal deep features to improve fetal health classification performance.	The approach involves complex deep learning and optimization processes that increase computational cost. The study also relies on a specific dataset and may require further validation on larger and more diverse clinical datasets for better generalization.
2	Jain E., Kaushik P., Kukreja V., Sakshi, Dogra A., Goyal B. (2025)	Vision Transformer (ViT)–based deep learning model applied to ultrasound imaging for fetal diagnostics and severity prediction.	Proposed a transformer-based framework to analyze ultrasound images for predicting fetal health conditions and severity levels, improving feature representation and classification accuracy compared to traditional CNN models.	The model requires large amounts of labeled ultrasound image data and high computational resources. Additionally, real-time clinical validation and testing across diverse populations are limited.
3	Rezende K. B. C., Bornia R. G., Rolnik D. L., Amim Jr. J., Ladeira L. P., Teixeira V. M. G., Cunha A. J. L. A. (2024)	External validation of the Fetal Medicine Foundation (FMF) competing risks model for predicting preeclampsia in the first trimester using maternal clinical characteristics and biomarkers.	Evaluated the effectiveness of the FMF competing risks model for early prediction of preeclampsia in a Brazilian population and assessed its predictive performance outside the original development population.	The study focuses on a specific regional population, which may limit generalizability to other populations. Additionally, prediction accuracy may vary depending on availability and quality of biomarkers and clinical data.
4	Yenkikar A., Singh V. K., Tamboli G., Charkha P., Bodke S., Bidwe R. V., Bali M. (2025)	Proposed a multi-modal AI framework integrating Siamese Neural Networks with Few-Shot Learning to analyze different fetal health data sources for early risk assessment.	The framework improves early fetal health prediction by combining multiple data modalities and enabling learning from limited labeled data, which enhances model adaptability in medical datasets.	The framework involves complex model architecture and may require careful tuning and integration of multiple data modalities. Its performance and reliability need further validation on larger clinical datasets and real-world healthcare environments.

Sl. No.	Author & Year	Method / Approach	Key Contribution	Gap Identified
5	Ettiyan R., Geetha V. (2026)	Proposed a Heap Optimization–based Multi-LSTM deep learning model integrated with wearable device data for continuous maternal and fetal health monitoring.	Developed a framework that utilizes wearable sensors and optimized LSTM networks to improve monitoring accuracy and enable continuous tracking of maternal and fetal health parameters.	The approach depends heavily on wearable device data quality and sensor reliability. Additionally, large-scale real-world testing and validation are required to ensure robustness and practical deployment in healthcare settings.
6	Abunadi I. (2025)	Applied machine learning techniques with feature engineering methods to analyze cardiotocography (CTG) signals for early detection of fetal anomalies.	Demonstrated that engineered CTG features combined with machine learning classifiers can improve early identification of abnormal fetal conditions and assist in fetal health monitoring.	The study primarily focuses on feature engineering and conventional machine learning models, which may limit the ability to capture complex patterns in CTG signals. Additionally, further validation on larger and more diverse datasets is required to ensure model robustness and generalizability.
7	Kapila R., Saleti S. (2023)	Proposed an ensemble machine learning model that combines feature selection and feature extraction techniques for analyzing cardiotocography (CTG) data.	Improved fetal health prediction performance by integrating multiple models and combining both feature selection and feature extraction to enhance classification accuracy.	Ensemble models can increase model complexity and computational cost. Additionally, performance evaluation may depend heavily on dataset characteristics, requiring further validation on diverse clinical datasets.
8	Zannah T. B., Tonni S. I., Sheakh M. A., Tahosin M. S., Sarower A. H., Begum M. (2025)	Conducted a comparative analysis of multiple ensemble learning algorithms (such as Random Forest, Gradient Boosting, and other ensemble techniques) for fetal health classification using CTG data.	Evaluated and compared the performance of different ensemble learning methods to identify the most effective model for fetal health classification.	The study mainly focuses on model comparison rather than system implementation. Further work is needed to integrate these models into practical healthcare systems and validate them with larger real-world clinical datasets.

Sl. No.	Author & Year	Method / Approach	Key Contribution	Gap Identified
9	Mushtaq G., Veningston K. (2024)	Proposed an AI-driven interpretable deep learning model for fetal health classification using cardiotocography (CTG) data, incorporating batch normalization and dropout layers to improve generalization and prevent overfitting, while providing interpretability for model decisions.	Achieved 99% accuracy and 93% sensitivity, outperforming traditional machine learning models such as Logistic Regression, KNN, SVM, Naïve Bayes, Random Forest, and Gradient Boosting. The model also provides interpretability, improving trust and usability in clinical decision-making.	Deep learning models require large training datasets and high computational resources. Further validation in real-world clinical environments is needed to ensure robustness and practical deployment.
10	Sirisha U., Srinivasu P. N., Padmavathi P., Kim S., Pavate A., Shafi J., Ijaz M. F. (2024)	Proposed a Dynamic Multi-Layer Perceptron (DMLP) model for fetal health classification using cardiotocography (CTG) data, where the neural network architecture dynamically adapts to improve learning and classification performance.	Demonstrated improved fetal health classification accuracy by leveraging adaptive neural network architecture to better capture complex relationships within CTG signals.	The model may require careful hyperparameter tuning and computational resources. Additionally, further validation using larger and diverse clinical datasets is necessary to confirm generalizability.
11	Francis F., Luz S., Wu H., Stock S. J., Townsend R. (2024)	Conducted a scoping review of machine learning techniques applied to cardiotocography (CTG) data for classifying fetal outcomes, analyzing different ML models, datasets, and evaluation methods used in previous studies.	Provided a comprehensive overview of existing machine learning approaches for fetal outcome classification using CTG signals and highlighted trends, performance comparisons, and research developments in this domain.	As a review study, it does not propose or implement a new predictive model. Additionally, variations in datasets, evaluation metrics, and experimental settings across studies make direct comparison of model performance challenging.
12	Subasi A., Kadasa B., Kremic E. (2020)	Applied a Bagging Ensemble Classifier to classify cardiotocography (CTG) data for predicting fetal health risks by combining multiple decision tree models to improve classification stability.	Demonstrated that the bagging ensemble approach improves prediction accuracy and robustness compared to single classifiers when analyzing CTG data for fetal risk detection.	Ensemble approaches may increase computational complexity and reduce model interpretability. Additionally, the study requires validation across larger and more diverse clinical datasets to confirm reliability.

Sl. No.	Author & Year	Method / Approach	Key Contribution	Gap Identified
13	Akbulut A., Ertugrul E., Topcu V. (2018)	Applied machine learning techniques to predict fetal health status using maternal clinical history data, analyzing various clinical parameters to classify fetal outcomes.	Demonstrated that maternal clinical history combined with machine learning algorithms can assist in predicting fetal health conditions and support early risk detection.	The study relies mainly on maternal clinical history data and may not capture physiological signals such as CTG data. Additional datasets and validation are required to improve prediction reliability and generalizability.
14	Melaet R., de Vries I. R., Kok R. D., Oei S. G., Huijben I. A. M., van Sloun R. J. G., van Laar J. O. E. H., Vullings R. (2024)	Developed an Artificial Intelligence–based method for cardiotocography (CTG) assessment during labor to predict fetal heart rate patterns and analyze prediction errors associated with abnormal CTG signals.	Demonstrated that the AI model can predict fetal heart rate patterns, and the prediction error correlates with FIGO-based clinical interpretation, indicating potential use as a clinical decision support tool during labor monitoring.	The method may fail in cases of abnormal CTG patterns, limiting reliability in high-risk scenarios. Further improvements and validation are required before large-scale clinical deployment.
15	Das S., Obaidullah S. M., Roy K., Saha C. K. (2023)	Evaluated the diagnostic performance of multiple machine learning algorithms for classifying fetal heart rate (FHR) baseline patterns using cardiotocography (CTG) data.	Conducted comparative analysis of several ML algorithms to determine their effectiveness in classifying fetal heart rate baseline patterns and improving fetal health monitoring.	The study mainly focuses on classification of FHR baseline patterns and may not consider other important CTG parameters. Further evaluation with comprehensive datasets and real-world clinical scenarios is required.
16	Abburi R., Hatai I., Jaros R., Martinek R., Arun Babu T., Arun Babu S., Samanta S. (2024)	Proposed an AI-based framework for remote fetal heart rate monitoring using wearable fetal phonocardiography (fPCG). The approach uses Chebyshev filtering and EC2EMDAN-PS-MODWT techniques for noise reduction, converts signals into spectrogram representations, and applies hybrid deep learning algorithms with optimization-based MA-DRL methods for classification.	Improved fetal heart rate monitoring by integrating advanced signal processing, spectrogram analysis, and hybrid deep learning, enabling more accurate remote monitoring and classification of fetal health signals.	The system depends on wearable sensor data quality and signal acquisition reliability. Deployment challenges and validation on larger real-world clinical datasets are still required.

Sl. No.	Author & Year	Method / Approach	Key Contribution	Gap Identified
17	Iraji M. S. (2019)	Proposed neural network–based models for fetal state prediction using cardiotocography (CTG) recordings and uterine activity signals. The study applied Deep Stacked Sparse Autoencoders and Deep-ANFIS (Adaptive Neuro-Fuzzy Inference System) with a multi-layer sub-ANFIS topology using 21 input variables from the UCI dataset.	Demonstrated that deep learning models provide higher precision in predicting fetal state compared to traditional approaches by effectively learning complex relationships in CTG and uterine activity signals.	The approach involves complex model architecture and high computational requirements. Additionally, validation using larger real-world clinical datasets is necessary to ensure generalization and practical clinical adoption.
18	Zhao Z., Deng Y., Zhang H., Zhao H., Lu C. (2023)	Proposed a CNN–RNN unified deep learning framework for classifying intrapartum cardiotocography (CTG) signals, where CNN extracts spatial features from CTG data and RNN captures temporal dependencies in fetal heart rate signals.	Demonstrated that combining CNN and RNN architectures improves the ability to capture both spatial and temporal characteristics of CTG signals, leading to better fetal health classification performance.	Deep learning models require large labeled datasets and significant computational resources. The study also needs further validation across diverse clinical datasets for broader applicability.
19	Ricciardi C., Improta G., Amato F., Cesarelli G., Romano M. (2020)	Applied machine learning techniques to classify delivery type using cardiotocographic (CTG) signals. A dataset of 370 CTG recordings was used, where 17 features were automatically extracted using custom software. Several ML algorithms including J48, Random Forest, Gradient Boosted Trees, AdaBoost (J48), and Decorate were evaluated using 10-fold cross-validation and SMOTE for class imbalance handling.	Demonstrated the effectiveness of machine learning models in analyzing CTG signals for predicting delivery type, providing comparative performance analysis using multiple evaluation metrics.	The study focuses mainly on delivery type classification rather than detailed fetal health risk prediction. Additionally, the dataset size is relatively limited and may require larger datasets for better generalization.
20	Sterckx L., Vandewiele G., Dehaene I., Janssens O., Ongenae F., De Backere F., De Turck F., Roelens K., Decruyenaere J., Van Hoecke S., Demeester T. (2020)	Proposed a clinical information extraction framework using Natural Language Processing (NLP) to analyze unstructured medical notes for predicting preterm birth risk. The approach utilizes biomedical ontologies, open-source tools, and active learning techniques to extract meaningful clinical features for predictive modeling.	Demonstrated that extracting features from unstructured clinical text data improves the interpretability and effectiveness of predictive models used in clinical decision support systems.	The approach depends heavily on the availability and quality of clinical text data. Variability in medical documentation and language complexity may affect extraction accuracy and model generalization.

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21	Lopes-Pereira J., Costa A., Ayres-De-Campos D., Costa-Santos C., Amaral J., Bernardes J. (2019)	Conducted an observational clinical study analyzing cardiotocography (CTG) and ST signals using computerized analysis systems across 38,466 deliveries to evaluate their impact on fetal monitoring and delivery outcomes.	Demonstrated that computerized CTG and ST signal analysis is associated with significant reductions in hypoxic-ischemic encephalopathy and cesarean deliveries, highlighting the clinical benefits of automated fetal monitoring systems.	The study is observational and focuses on clinical outcome association rather than developing or evaluating advanced predictive AI models. Further research is required to integrate more advanced machine learning or deep learning approaches for predictive analysis.
22	Lin X., Liang C., Liu J., Lyu T., Ghumman N., Campbell B. (2024)	Conducted a systematic review of Artificial Intelligence–augmented Clinical Decision Support Systems (CDSS) used in pregnancy care, analyzing different AI techniques applied for maternal and fetal health monitoring.	Provided a comprehensive overview of how AI-based decision support systems assist clinicians in pregnancy care, highlighting the potential of AI for improving diagnostic accuracy, monitoring, and clinical decision-making.	As a review study, it does not propose a new predictive model. Additionally, variability in datasets, AI techniques, and evaluation methods across studies makes direct comparison and standardization difficult.
23	Wu S., Qiu S., Ma J., Song X., Wu H., Wang B. (2023)	Proposed a multifactorial prediction model combining Color Doppler Ultrasound parameters and Fetal Heart Rate (FHR) monitoring data to detect early signs of fetal distress.	Demonstrated that integrating multiple physiological monitoring methods improves early detection of fetal distress and enhances predictive capability compared to using a single monitoring technique.	The model depends on multiple clinical data sources, which may not always be available in all healthcare settings. Additionally, further validation with larger patient datasets is required to confirm reliability and clinical applicability.
24	Zhang J., Xiao S., Zhu Y., Zhang Z., Cao H., Xie M., Zhang L. (2024)	Reviewed the advances in Artificial Intelligence applications in fetal echocardiography, focusing on deep learning techniques for medical image preprocessing, feature extraction, and disease diagnosis in fetal heart imaging.	Highlighted how AI and deep learning improve fetal echocardiography analysis, enabling better biometric measurement, congenital heart disease (CHD) detection, and automated image interpretation.	Most AI models require large labeled medical imaging datasets and high computational resources. Additionally, further validation and integration into routine clinical workflows are needed for practical adoption.
25	Mursil M., Rashwan H. A., Khalid A., Cavallé-Busquets P., Santos-Calderon L., Murphy M. M., Puig D. (2025)	Proposed an interpretable deep neural network model for predicting neonatal birth weight (BW) using multimodal maternal factors such as nutrition, physiological parameters, and lifestyle information during early pregnancy.	Demonstrated that integrating multiple maternal factors with deep learning models improves early prediction of low birth weight (LBW). The model also provides interpretability, helping clinicians identify fetuses at risk.	The model relies on availability and accuracy of maternal health data, which may vary across populations. Further validation with larger and diverse clinical datasets is required for broader applicability.

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Emerging Diagnostic/Detecting Approaches

• Cardiotocography (CTG) signals and ultrasound images are increasingly analyzed using deep learning techniques, including CNN, RNN, LSTM, and Transformer models, to ensure precise fetal health predictions.

• In order to enhance the efficacy of feature selection and classification, hybrid models that integrate deep learning with optimization algorithms (e.g., War Strategy Optimization, Whale Optimization) are implemented.

• Architectures based on Vision Transformer (ViT) are becoming increasingly popular for sophisticated ultrasound image analysis, as they provide superior feature representation in comparison to conventional CNNs.

• In order to improve the accuracy of predictions, multimodal AI frameworks incorporate a variety of data sources, including ultrasound images, maternal clinical data, CTG signals, and Doppler parameters.

• In order to effectively manage limited labeled medical datasets, few-shot learning approaches are being investigated.

• Continuous real-time surveillance of maternal and fetal health is facilitated by wearable sensor-based systems.

• Home-based healthcare solutions are increasingly incorporating remote monitoring techniques that utilize fetal phonocardiography (fPCG) and IoT-based systems.

- Random Forest, Gradient Boosting, and Bagging are popular ensemble learning methods that are employed to enhance the robustness and stability of classification.
- Clinical Decision Support Systems (CDSS) that are AI-based aid clinicians in making informed decisions during pregnancy care.
- Natural Language Processing (NLP) techniques are employed to extract valuable features from unstructured clinical notes.
- Artificial intelligence is coupled with advanced signal processing techniques (e.g., spectrogram analysis, filtering methods) to enhance feature extraction.
- AI approaches that are comprehensible are being developed to enhance the trust and transparency of medical predictions.

Identified Trends and Achievements

The transition from conventional machine learning methods to advanced deep learning and hybrid AI models is evident.

- Some models have attained exceptionally high performance levels, and there have been substantial enhancements in prediction accuracy and sensitivity.
- The capacity to automatically extract intricate features from medical images and CTG signals has been improved by deep learning models.
- The reliability and robustness of fetal health prediction systems have been enhanced through the integration of multimodal data.
- The integration of ubiquitous devices and IoT has made real-time monitoring and remote healthcare solutions more feasible.
- The significance of interpretable AI models is increasing, as they assist clinicians in comprehending the decisions made by the models and foster a greater sense of confidence in AI systems.
- The necessity for manual intervention has been diminished as a result of automated feature engineering and optimization techniques.
- The early detection of abnormalities has been enhanced by AI implementations in fetal echocardiography and ultrasound imaging.
- Artificial intelligence (AI) is being progressively integrated into clinical decision support systems to facilitate risk assessment and diagnosis.

Persistent Gaps and Challenges

- The deployment of deep learning and hybrid models in real-time clinical environments is frequently challenging due to the high computational resources they necessitate.
- The medical domain is characterized by privacy and data availability concerns, which restrict the availability of large labeled datasets, which are the foundation of the majority of models.
- The real-world clinical validation and large-scale deployment of proposed models are still limited.

• The quality and reliability of input data, including CTG signals and wearable sensor data, have a significant impact on the efficacy of models.
• The complexity of the system is increased by the integration of multiple data modalities, necessitating meticulous design and synchronization.
• It is challenging to conduct a fair comparison of models due to the variations in datasets, preprocessing methodologies, and evaluation metrics.
• Infrastructure constraints present deployment challenges in low-resource healthcare environments.

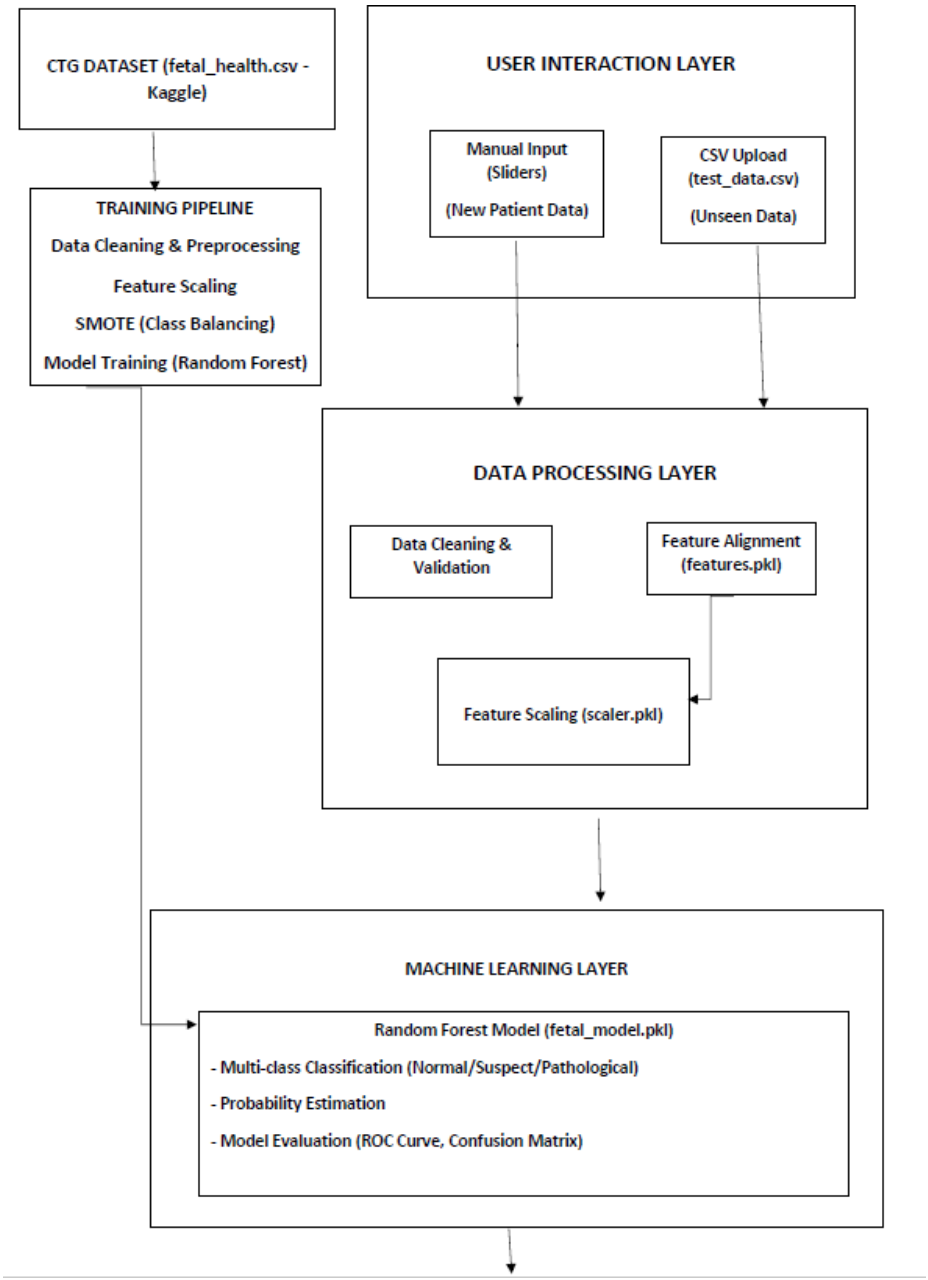
Consolidated Summary

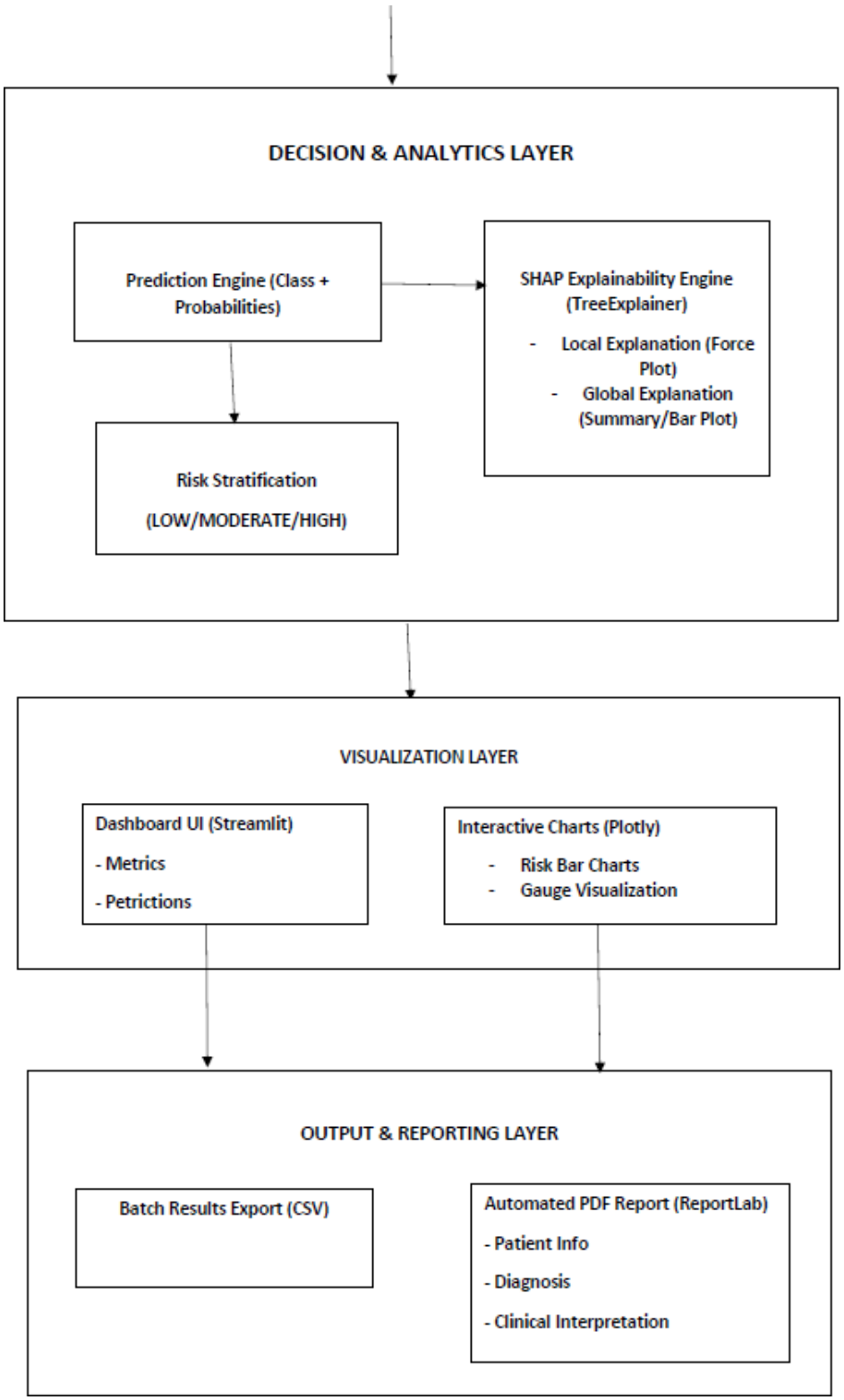
Recent developments in fetal health monitoring have underscored a substantial transition to data-centric and AI-driven methodologies, particularly through the implementation of multimodal frameworks and deep learning. By utilizing a variety of data sources, including ultrasound imaging, maternal clinical information, and CTG signals, these methods have significantly enhanced the accuracy and efficacy of fetal distress detection and early risk prediction. Furthermore, emerging technologies prioritize the integration of interpretable AI techniques, real-time monitoring, and remote healthcare through wearable devices to improve clinical trust and decision-making. While these advancements have been made, there are still numerous obstacles to overcome, such as the limited availability of diverse and large datasets, the high computational demands, the absence of generalization across populations, and the insufficient actual clinical validation. As a result, it is anticipated that future research will concentrate on the development of systems that are scalable, explainable, and robust, and that can be effectively deployed in practical healthcare environments while maintaining reliability and accessibility.

Chapter 3
Architecture Diagram

The overall architecture of the proposed system integrates multiple components from data preprocessing through model fusion, interpretability, and reporting to facilitate cancer detection in histopathological images. Below is a detailed explanation of each block in the system's architecture:

Fig 3.1 System Architecture




DECISION & ANALYTICS LAYER
Prediction Engine (Class + Probabilities)
SHAP Explainability Engine (TreeExplainer)
- Local Explanation (Force Plot)
- Global Explanation (Summary/Bar Plot)
Risk Stratification
(LOW/MODERATE/HIGH)
VISUALIZATION LAYER
Dashboard UI (Streamlit)
- Metrics
- Petrictions
Interactive Charts (Plotly)
- Risk Bar Charts
- Gauge Visualization
OUTPUT & REPORTING LAYER
Batch Results Export (CSV)
Automated PDF Report (ReportLab)
- Patient Info
- Diagnosis
- Clinical Interpretation

The proposed system follows a layered architecture that integrates data processing, machine learning, explainable AI, and clinical reporting to provide accurate and interpretable fetal health assessment.

1. CTG Dataset (fetal_health.csv – Kaggle)

- The system is initially trained using the Cardiotocography (CTG) dataset.
- This dataset contains multiple fetal health parameters such as:
  - Baseline fetal heart rate
  - Variability measures
  - Accelerations and decelerations
- Each record is labeled into three classes:
  - Normal
  - Suspect
  - Pathological
- The dataset is used only during the training phase to build the machine learning model.
- It forms the foundation for learning clinical patterns associated with fetal conditions.

2. Training Pipeline

This stage is responsible for building the predictive model.

- Data Cleaning & Preprocessing: Removes inconsistencies and ensures data quality.
- Feature Scaling: Standardizes numerical values using techniques like StandardScaler.
- SMOTE (Synthetic Minority Oversampling Technique): Balances class distribution by generating synthetic samples for minority classes. Helps improve model performance on underrepresented classes.
- Model Training**:** A Random Forest Classifier is trained on the processed dataset. Learns complex relationships between CTG features and fetal health conditions.
- The trained model is saved as:
  - `fetal_model.pkl`
  - `scaler.pkl`
  - `features.pkl`

3. User Interaction Layer

- This layer enables interaction between the user and the system.
- Provides two input methods:
  - Manual Input (Sliders):
    Allows clinicians to input real-time patient data.
  - CSV Upload (Batch Processing):
    Enables analysis of multiple patients simultaneously. Ensures flexibility for both single-patient and bulk analysis**.**

4. Data Processing Layer

Prepares input data before feeding it into the model.

- Data Cleaning & Validation: Ensures input values are within valid clinical ranges.
- Feature Alignment**:** Matches input data with the trained model's feature structure using `features.pkl`.
- **Feature Scaling**: Applies the same scaling used during training (`scaler.pkl`).

Ensures consistency between training data and real-time input data.

5. Machine Learning Layer

- Core predictive component of the system.
- Uses the trained Random Forest model (`fetal_model.pkl`).
- Performs:
  - Multi-class classification: Normal / Suspect / Pathological
  - Probability estimation: Outputs likelihood for each class Includes model evaluation metrics:
  - Accuracy
  - ROC Curve
  - Confusion Matrix
- Ensures reliable and validated predictions.

6. Decision & Analytics Layer

Converts model outputs into meaningful clinical insights.

Prediction Engine

Generates:

- Final predicted class
- Class-wise probabilities

Risk Stratification

Converts probabilities into risk levels:

- LOW (Normal)
- MODERATE (Suspect)
- HIGH (Pathological)

Helps clinicians quickly assess severity.

SHAP Explainability Engine

- Provides model interpretability using SHAP (TreeExplainer).
- Two types of explanations:
  - **Local Explanation**: Explains prediction for a single patient (Force Plot)
  - **Global Explanation**: Shows overall feature importance (Summary/Bar Plot)
- Enhances trust and transparency in AI decisions.

7. Visualization Layer

- Presents outputs in an interactive and user-friendly format.
- Built using Streamlit and Plotly.
- Includes:
  - Dashboard interface
  - Prediction summaries
  - Risk indicators
  - Interactive charts:
    - Risk distribution bar charts
    - Gauge visualization
- Helps users interpret results visually.

8. Output & Reporting Layer

- Generates final outputs for clinical use.

Batch Results Export

- Allows downloading results as CSV files.
- Useful for hospital record management.

Automated PDF Report

- Generates a structured clinical report using ReportLab.
- Includes:
  - Patient information
  - CTG parameters
  - AI diagnosis
  - Probability distribution
  - Clinical interpretation
- Designed to be hospital-grade and documentation-ready.

Chapter 4

Code Snippets

4.1 CODE SNIPPETS

1. Dataset Loading and Initial Preprocessing

Loads the fetal health dataset and prepares features and target variable for training.

```
import pandas as pd

# Load dataset
df = pd.read_csv("fetal_health.csv")

# Map target labels
label_map = {1.0: 0, 2.0: 1, 3.0: 2}
df["target"] = df["fetal_health"].map(label_map)

# Split features and target
X = df.drop(["fetal_health", "target"], axis=1)
y = df["target"]
```

2. Model Loading and SHAP Initialization

Loads trained model, scaler, features, and initializes SHAP explainer.

```
@st.cache_resource
def load_models():
    model = joblib.load("fetal_model.pkl")
    scaler = joblib.load("scaler.pkl")
    features = joblib.load("features.pkl")
    classes = ["Normal", "Suspect", "Pathological"]

    explainer = shap.TreeExplainer(model)
    return model, scaler, features, classes, explainer

model, scaler, features, classes, explainer = load_models()
```

3. Global SHAP Analysis

Displays overall feature importance across dataset using SHAP.

```
shap_values = explainer.shap_values(X_test)

shap.summary_plot(
    shap_values,
    X_test,
    plot_type="bar"
)
```

4. Batch Prediction

Processes multiple patient records and predicts fetal health.

```
df = pd.read_csv(uploaded_file)
X = scaler.transform(df)

preds = model.predict(X)
probs = model.predict_proba(X)
```

5. Single Patient Prediction

Generates prediction for manually entered CTG values.

```
patient = dict.fromkeys(features, 0.0)
patient["baseline value"] = baseline
patient["abnormal_short_term_variability"] = variability

df_pt = pd.DataFrame([patient])
X_pt = scaler.transform(df_pt)

prediction = model.predict(X_pt)
```

6. SHAP Local Explanation

Explains individual prediction using SHAP force plot.

```
shap_values = explainer.shap_values(df_pt)

st_shap(shap.force_plot(
    explainer.expected_value[0],
    shap_values[0],
    df_pt
))
```

7. Confusion Matrix

Evaluates classification performance.

```
cm = confusion_matrix(y_test, y_pred)

sns.heatmap(cm, annot=True, fmt="d")
plt.show()
```

8. ROC Curve

Shows model performance using ROC-AUC.

```
fpr, tpr, _ = roc_curve(y_test_bin[:,0], y_prob[:,0])
plt.plot(fpr, tpr)
plt.show()
```

9. Model Training (Random Forest)

Trains the fetal health prediction model.

```
model = RandomForestClassifier(n_estimators=200)
model.fit(X_train, y_train)
```

10. SMOTE for Data balancing

Balances dataset to handle class imbalance.

```
smote = SMOTE()
X_res, y_res = smote.fit_resample(X_train, y_train)
```

11. SHAP Beeswarm (Distribution Plot)

Displays feature impact distribution for all samples.

```
st_shap(shap.summary_plot(
    shap_values_val,
    val_data["X_test"],
    plot_type="bar",
    show=False
))
```

12. Risk Level Mapping Logic

Maps prediction output to clinical risk categories.

```
risk_levels = {
    "Normal": "LOW",
    "Suspect": "MODERATE",
    "Pathological": "HIGH"
}

risk_level = risk_levels[prediction]
```

13. Gauge Chart for Risk Visualization

Visualizes pathological risk percentage in an intuitive format.

```
fig = go.Figure(go.Indicator(
    mode="gauge+number",
    value=risk*100,
    gauge={"axis": {"range": [0,100]}}
))

st.plotly_chart(fig)
```

14. Feature Importance Extraction (Top Factors)

Extracts most important features from trained model.

```
imp_df = pd.DataFrame({
    "Feature": features,
    "Importance": model.feature_importances_
}).sort_values("Importance", ascending=False)
```

15. PDF Report Generation

Generates automated hospital-style fetal health report.

```
def generate_clinical_report(patient_data, prediction):
    doc = SimpleDocTemplate("report.pdf", pagesize=A4)
    elements = []

    elements.append(Paragraph("Fetal Health Report", getSampleStyleSheet()["Title"]))
    elements.append(Paragraph(f"Diagnosis: {prediction}", getSampleStyleSheet()["Normal"]))

    doc.build(elements)
```

Chapter 5

Results and Discussion

5.1. TRAINING PHASE RESULTS

The model was trained using a Random Forest classifier with SMOTE-based class balancing to address data imbalance in fetal health categories. The learning behavior was stable, and the training process obtained strong convergence. The homogenous contribution of all input variables was guaranteed through the use of StandardScaler for feature scaling. The final trained model obtained an overall accuracy of approximately 91% on the test dataset, suggesting that the CTG patterns were effectively learned for the classification of fetal health.

5.2. VALIDATION AND INFERENCE PERFORMANCE

Confusion Matrix Analysis

The confusion matrix demonstrates the classification performance across three classes: Normal, Suspect, and Pathological.

- **Normal Class:**

Correctly classified: 307 out of 332 cases (~92%)
Misclassified mainly as Suspect (24 cases), indicating slight overlap between normal and borderline patterns.

- **Suspect Class:**

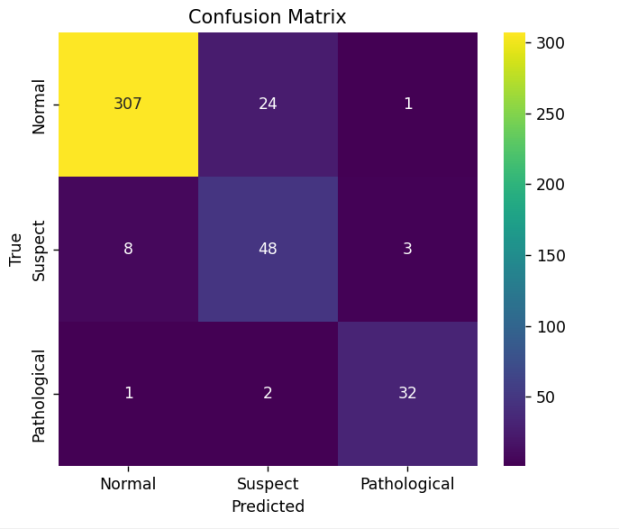
Correctly classified: 48 out of 59 cases (~81%)
Some misclassification into Normal and Pathological, showing inherent ambiguity in this intermediate class.

- **Pathological Class:**

Correctly classified: 32 out of 35 cases (~91%)
Very few misclassifications, indicating strong detection of high-risk cases.

Insight:
The model performs best for Normal and Pathological classes, while Suspect remains challenging, which aligns with real-world clinical uncertainty.

Fig 5.1 Confusion Matrix



Classification Report Analysis

- **Precision:** High for Normal (0.97) and Pathological (0.89), lower for Suspect (0.65)
- **Recall:** Strong across all classes, especially Pathological (0.91)
- **F1-score:** Balanced performance with overall weighted average of **0.91**

Interpretation:
The model is highly reliable in detecting critical (Pathological) cases, which is crucial for clinical safety.

Fig 5.2 Classification Report

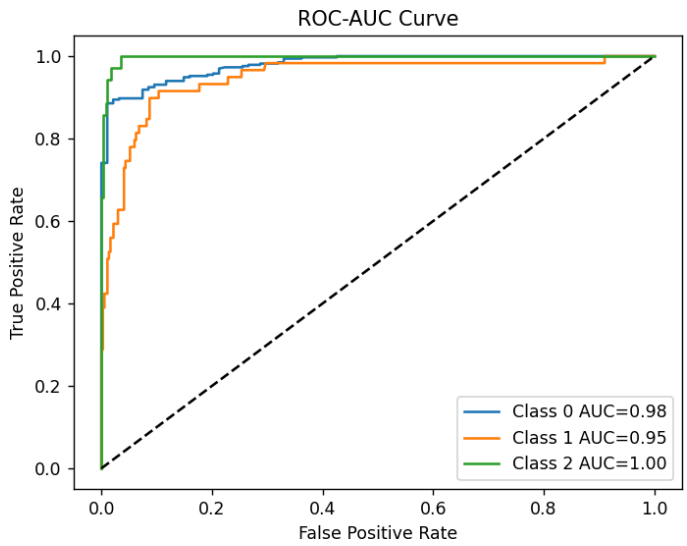
```
Classification Report:

              precision    recall  f1-score   support

           0       0.97      0.92      0.95       332
           1       0.65      0.81      0.72        59
           2       0.89      0.91      0.90        35

    accuracy                           0.91       426
   macro avg       0.84      0.88      0.86       426
weighted avg       0.92      0.91      0.91       426
```

ROC-AUC Curve Analysis

- Class 0 (Normal): **AUC = 0.98**
- Class 1 (Suspect): **AUC = 0.95**
- Class 2 (Pathological): **AUC = 1.00**

Insight:
The near-perfect AUC for Pathological class indicates excellent separability, meaning the model can confidently distinguish high-risk cases.

Fig 5.3 ROC Curve

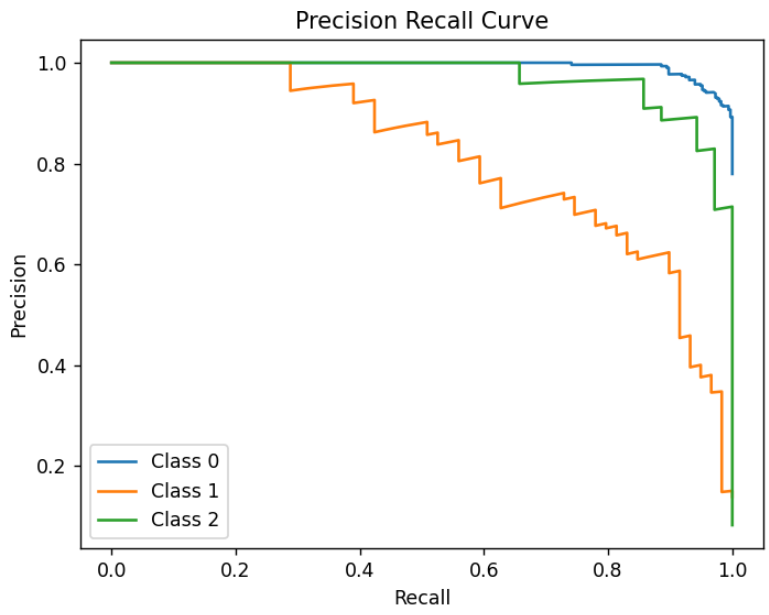


Precision-Recall Curve Analysis

- High precision maintained for Normal and Pathological classes across recall values
- Suspect class shows variability, especially at higher recall

Interpretation:
This confirms that the model maintains high reliability in critical predictions, while borderline cases remain complex.

Fig 5.4 Precision-Recall Curve



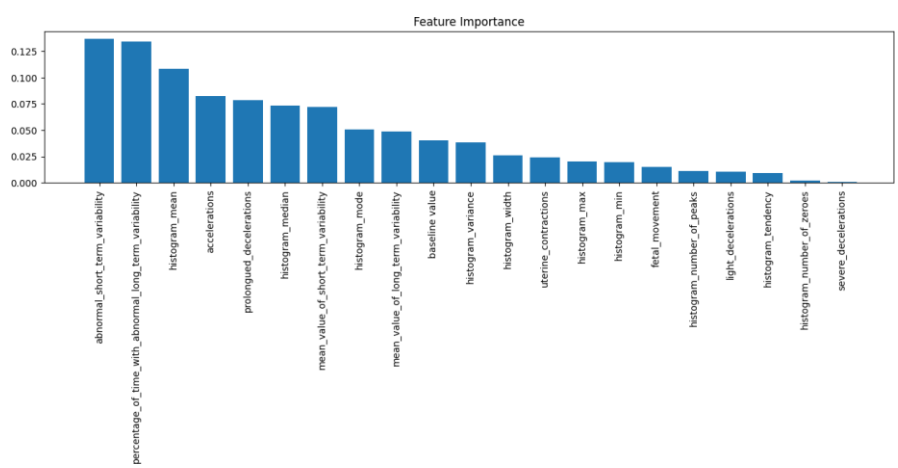
Feature Importance Analysis

The feature importance plot highlights key contributors:

- Abnormal short-term variability
- Percentage of abnormal long-term variability
- Histogram mean
- Accelerations
- Prolonged decelerations

Insight:
These features are clinically relevant indicators of fetal distress, validating the model's alignment with medical knowledge.

Fig 5.5 Feature Importance



5.3. SHAP VISUALIZATION FINDINGS

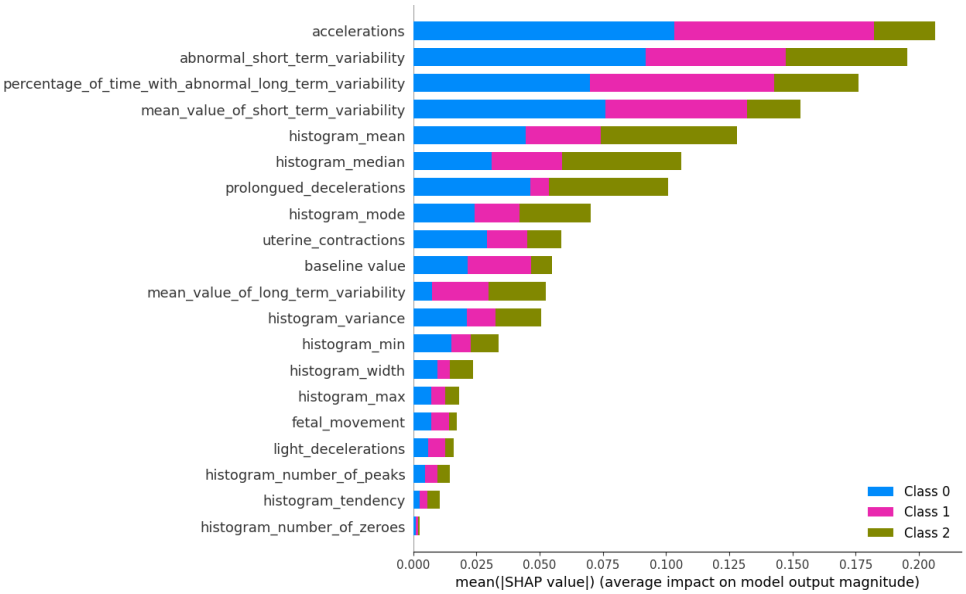
Global SHAP Analysis

The SHAP bar plot shows:

- Features like accelerations, variability metrics, and decelerations dominate model decisions
- Different classes are influenced by different feature patterns

Interpretation:
The model does not rely on a single feature but uses a combination of physiological indicators**,** improving robustness.

Fig 5.5 Global SHAP Visualizations



SHAP Beeswarm Analysis

- Displays distribution of feature impact across all samples
- High variability features show both positive and negative contributions
- Clear separation in SHAP values for different classes

Insight:

- High abnormal variability → pushes prediction toward Pathological
- High accelerations → pushes toward Normal

This matches real-world fetal monitoring interpretations.

Fig 5.6 SHAP Beeswarm Analysis

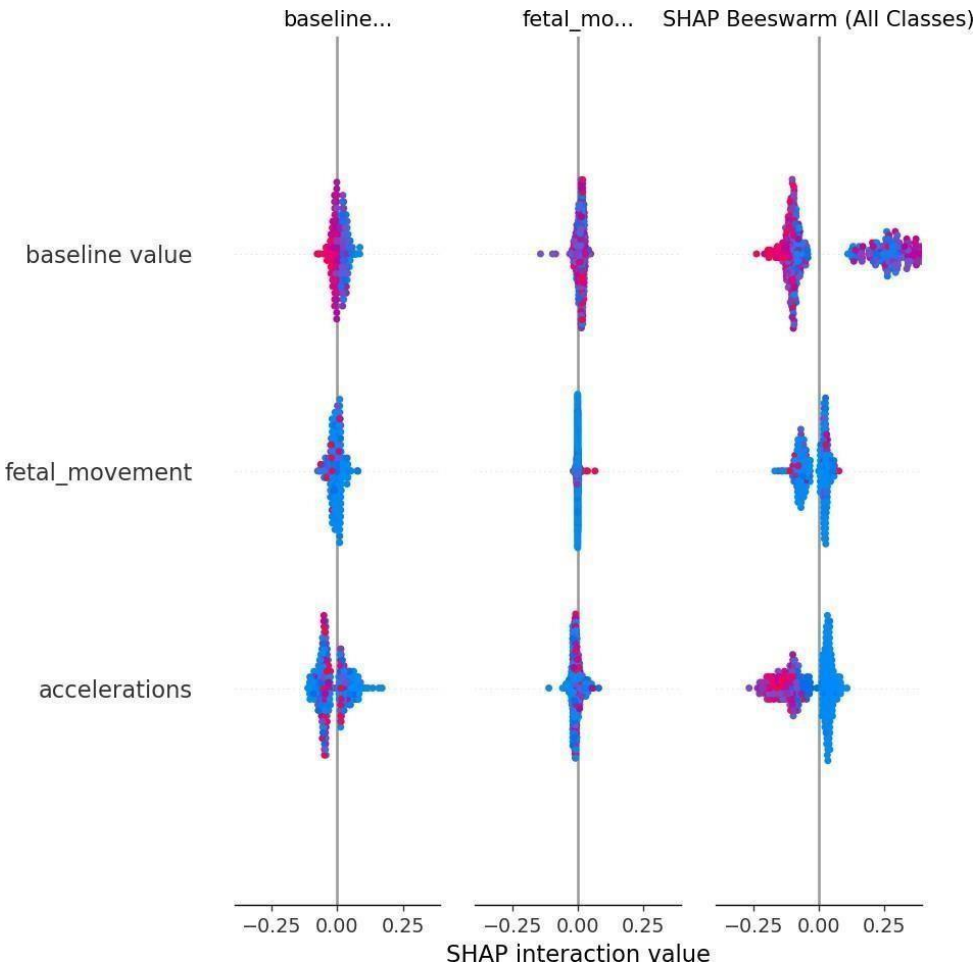


Fig 5.7 Streamlit Dashboard

A	B	C	D	E
16	Normal	2.70%	LOW	
17	Normal	2.40%	LOW	
18	Normal	0.20%	LOW	
19	Normal	4.20%	LOW	
20	Normal	0.10%	LOW	
21	Normal	0.80%	LOW	
22	Normal	2.30%	LOW	
23	Normal	13.90%	LOW	
24	Normal	0.10%	LOW	
25	Normal	0.10%	LOW	
26	Normal	2.50%	LOW	
27	Normal	0.10%	LOW	
28	Normal	5.70%	LOW	
29	Suspect	2.40%	MODERATE	
30	Suspect	4.30%	MODERATE	
31	Suspect	14.30%	MODERATE	
32	Normal	0.10%	LOW	
33	Normal	1.90%	LOW	
34	Normal	0.10%	LOW	
35	Pathologic	76.00%	HIGH	
36	Normal	2.10%	LOW	
37	Normal	0.70%	LOW	
38	Suspect	13.70%	MODERATE	
39	Normal	0.10%	LOW	
40	Normal	2.00%	LOW	
41	Normal	0.70%	LOW	
42	Normal	18.90%	LOW	
43	Normal	0.00%	LOW	
44	Normal	0.10%	LOW	
45	Normal	0.10%	LOW	

Fetal Health Assessment Report

Cardiotocography (CTG) Analysis

PATIENT INFORMATION

Patient ID: 001 | **Date:** 19/03/2026 15:19

CTG PARAMETERS

Parameter	Measured	Normal Range
Baseline FHR (bpm)	140.00	110-160
Abnormal STV (%)	20.00	<50%
Prolonged Decels	0.00	0-0.1
Accelerations	0.00	>0.005
Light Decels	0.00	<0.01

AI FETAL HEALTH CLASSIFICATION

Primary Diagnosis: PATHOLOGICAL

Probability Distribution:
- Normal: **40.2%**
- Suspect: **15.4%**
- Pathological: **44.4%**

CLINICAL INTERPRETATION

PATHOLOGICAL FETAL STATUS

CRITICAL CTG abnormalities confirmed. Pathological risk: **44.4%**

URGENT MANAGEMENT:
1. URGENT OBSTETRIC REVIEW
2. Stop oxytocin infusion
3. Left lateral position
4. 500ml IV fluids stat
5. Prepare for delivery if no improvement

KEY DECISION FACTORS

Rank	Feature	Model Weight
8	abnormal_short_term_variability	0.137
10	percentage_of_time_with_abnormal_long_term_variability	0.134
18	histogram_mean	0.108
2	accelerations	0.083
7	prolongued_decelerations	0.079

CTG FEATURE CLINICAL MAPPING

Feature	Clinical Meaning	High SHAP → Risk
abnormal_short_term_variability	Reduced FHR variability	Pathological (distress)
prolongued_decelerations	Late/variable decels	Suspect (hypoxia)
accelerations	Reactive FHR response	Normal (healthy)
baseline value	Mean FHR (bpm)	Extreme → Pathological

DISCLAIMER: AI supports clinical decision making per NICE/IPOC guidelines. Final decisions by attending clinician.

5.4. QUANTITATIVE REPORT INTERPRETATION

The cardiotocography (CTG) analysis suggests the presence of significant abnormalities in fetal heart rate patterns, indicating a pathological fetal state. An elevated risk of fetal distress necessitating immediate clinical attention is indicated by the pathological probability of 44.4% assigned by the model.

The fetal heart rate is within the normal range at the baseline; however, the aberrant classification is influenced by the absence of accelerations and the presence of concerning variability patterns. These results may suggest that fetal oxygenation is compromised or that autonomic responsiveness is reduced, both of which are clinically associated with adverse fetal outcomes.

The model's decision has been substantially influenced by key contributing factors, such as abnormal short-term variability and reduced accelerations, from an explainability perspective. The model's prediction is in accordance with the standard obstetric assessment guidelines, as these features are well- established clinical indicators of fetal compromise.

The condition cannot be regarded as reassuring in light of the current assessment. It is advisable to conduct an immediate clinical evaluation in order to verify the fetal status and avert any potential deterioration.

5.5. RECOMMENDED CLINICAL ACTIONS

- Initiate continuous CTG monitoring to observe evolving patterns
- Reposition the mother to left lateral position to improve uteroplacental blood flow
- Administer intravenous fluids to enhance maternal circulation
- Discontinue any uterotonic agents (e.g., oxytocin) if in use
- Consider further diagnostic interventions such as fetal scalp stimulation or blood sampling
- Prepare for timely obstetric intervention if no improvement is observed

5.6. SYSTEM EVALUATION

The developed system integrates:

- Real-time prediction
- SHAP-based explainability
- Batch processing capability
- Automated clinical report generation

Performance Characteristics:

- Fast inference (real-time prediction)
- Interpretable outputs (SHAP explanations)
- Scalable for multi-patient analysis

Table 5.1 Performance Benchmarks

Metric	Value
Accuracy	91%
ROC-AUC (Normal)	0.98
ROC-AUC (Suspect)	0.95
ROC-AUC (Pathological)	1.00
Precision (Weighted)	0.92
Recall (Weighted)	0.91
F1-score (Weighted)	0.91

Discussion:

The performance benchmark results indicate that the proposed model achieves a strong balance between accuracy and reliability, with an overall accuracy of 91% and high ROC-AUC scores across all classes. The near-perfect AUC for the Pathological class highlights the model's effectiveness in identifying critical fetal conditions, which is essential in clinical scenarios. Additionally, the consistency between precision, recall, and F1-score values demonstrates stable and well-generalized model performance. Overall, these metrics confirm that the system is both accurate and dependable for real-world fetal health assessment.

5.7. ISSUE ANALYSIS AND SOLUTIONS

Table 5.2

Issue	Observed Impact	Applied Solution
Class Imbalance in Dataset	Majority class (Normal) dominated training, leading to biased predictions and reduced performance for Suspect and Pathological classes	Applied SMOTE (Synthetic Minority Oversampling Technique) to balance class distribution and improve minority class learning
Lower Precision for Suspect Class	Increased misclassification between Normal and Pathological classes due to overlapping feature patterns	Used ensemble model (Random Forest) and improved feature representation to better capture complex decision boundaries
Overlapping Clinical Features	Similar CTG patterns between Normal and Suspect cases reduced classification clarity	Incorporated multiple feature interactions and leveraged SHAP to understand feature contributions for better model tuning
Computational Cost of SHAP	High latency during real-time prediction due to SHAP value computation	Implemented pre-computation of SHAP values on validation data to reduce runtime overhead in dashboard
Model Interpretability Challenges	Black-box nature of ML models reduces trust in clinical settings	Integrated SHAP explainability (global + local) to provide transparent and interpretable predictions
Risk of Overfitting	Complex feature relationships may lead to overfitting on training data	Used Random Forest with controlled depth and multiple estimators to ensure generalization
Real-time Deployment Constraints	Need for fast and scalable predictions in clinical environments	Optimized pipeline using feature scaling, caching, and efficient inference mechanisms
Lack of Clinical Context in Raw Outputs	Model predictions alone are not sufficient for clinical decision-making	Developed automated PDF reports with clinical interpretation and recommendations
Handling Multi-patient Data	Difficulty in analyzing large datasets manually	Implemented batch processing via CSV upload for scalable multi- patient analysis

Result:
These improvements enhanced both model accuracy and usability in real-world scenarios.

5.8. DISCUSSION SUMMARY

The experimental results demonstrate that the proposed system effectively combines machine learning and explainable AI to achieve high accuracy and reliability in fetal health classification. The model performs exceptionally well in identifying Normal and Pathological conditions, while the Suspect class presents moderate challenges due to its transitional nature. The integration of SHAP enhances transparency by providing clear explanations for model predictions, thereby improving clinical trust. Overall, the system achieves a strong balance between performance, interpretability, and practical usability.

5.9. FINAL OUTCOME

The developed system successfully delivers an intelligent fetal health monitoring solution capable of accurate prediction, real-time analysis, and interpretable decision-making. With its strong performance metrics, explainability features, and automated reporting capabilities, the system demonstrates significant potential for real-world clinical deployment, assisting healthcare professionals in early detection and management of fetal distress.

Chapter 6

Practical and Clinical Implications

6.1. CLINICAL RELEVANCE

The intelligent and dependable solution of the proposed fetal health monitoring system for the analysis of cardiotocography (CTG) data demonstrates its significant clinical relevance. In obstetrics, the early detection of fetal distress is essential, as delayed intervention can result in severe complications. The model that has been devised is highly accurate in its ability to identify pathological cases, thereby preventing the neglect of high-risk conditions.

The integration of SHAP (SHapley Additive Explanations) is a critical asset of the system, as it improves interpretability. In contrast to conventional black- box models, this system offers explicit explanations for each prediction by emphasizing the role of individual features, including embryonic heart rate variability, accelerations, and decelerations. This is in close alignment with the manner in which clinicians interpret CTG data, thereby enhancing the trust and efficacy of medical settings.

Additionally, the system functions as a clinical decision-support instrument, assisting physicians rather than substituting them. It assures consistent analysis across patients, minimizes subjective variability in interpretation, and reduces manual workload. The system is particularly well-suited for real-world clinical environments, as it can offer both predictions and explanations, which is particularly beneficial for less experienced practitioners.

6.2. INTEGRATION WITH HOSPITAL INFORMATION SYSTEMS (HIS)

In order to facilitate seamless workflow integration, the proposed system is intended to be readily integrated into existing Electronic Health Record (EHR) and Hospital Information Systems (HIS) platforms. The system can directly analyze and predict CTG data collected from monitoring devices in real time.

The system can interact with hospital databases and software platforms through API-based communication, which enables integration. The system is capable of supporting the following:

- The direct retrieval of patient CTG data from hospital databases through automated data ingestion

- Real-time prediction and alert generation, particularly for high-risk (pathological) cases
- Preservation of SHAP explanations and prediction results for future clinical reference and auditing

Furthermore, the system has the capability to produce automated diagnostic reports in PDF format, which can be recorded in the patient's medical record. This enhances documentation and guarantees the traceability of decisions.

The system is scalable and lightweight from a deployment perspective, rendering it appropriate for:

- Large facilities with a digital infrastructure that is integrated
- Clinics that are small and have limited resources
- Cloud-based deployments for remote healthcare settings

This integration improves clinical efficiency, reduces diagnostic delays, and facilitates data-driven decision-making.

6.3. POTENTIAL MARKET OR COMMERCIAL IMPACT

The system has significant market potential within the rapidly growing field of AI-driven healthcare technologies. With increasing emphasis on early diagnosis, preventive care, and cost-effective solutions, there is a strong demand for intelligent medical systems.

Target Market Segments

- Hospitals and maternity care centers
- Diagnostic laboratories handling CTG data
- Telemedicine and remote healthcare platforms
- Rural healthcare centers with limited specialist availability

Commercialization Opportunities

The system can be developed into:

- A **Clinical Decision Support System (CDSS)** integrated into hospital software
- A **Software-as-a-Service (SaaS) platform** for remote fetal monitoring
- A **cloud-based AI diagnostic tool** accessible through web applications

Competitive Advantage

The inclusion of SHAP-based explainability provides a major advantage over traditional AI systems, as regulatory bodies and healthcare providers increasingly demand transparency in decision-making. Additionally, the system's ability to generate automated reports and handle batch data processing makes it practical for real-world deployment.

Future enhancements such as IoT-based real-time monitoring and mobile integration can further expand its market reach, making it a scalable and commercially viable solution.

6.4. VISUALIZATION OF MODEL FAILURE CASES

Table 6.1 Error Type, Observation and Cause

Error Type	Observation	Cause
Normal misclassified as Suspect	Cases with generally normal patterns but slight deviations in variability or occasional irregularities were predicted as Suspect	Overlapping feature values between Normal and Suspect classes, especially in variability-related features
Suspect misclassified as Normal	Some borderline cases were predicted as Normal despite mild abnormal indicators	Weak expression of abnormal features, leading the model to favor dominant Normal class patterns
Suspect misclassified as Pathological	Certain Suspect cases with slightly higher abnormal values were classified as Pathological	Presence of strong feature signals (e.g., variability or decelerations) resembling critical conditions
Pathological misclassified as Suspect	A few high-risk cases were predicted as Suspect instead of Pathological	Less pronounced abnormal patterns or missing extreme feature values
Feature Contribution Misinterpretation	SHAP values in some cases showed mixed positive and negative contributions, causing ambiguity in prediction	Complex interaction between multiple features influencing the final decision
Borderline Case Confusion	Samples near class boundaries showed inconsistent predictions	Intrinsic uncertainty in CTG data where clinical distinction itself is difficult

Key Insight

The majority of errors occur in the Suspect class, which is expected due to its intermediate nature. The analysis confirms that most misclassifications arise from overlapping feature distributions and subtle variations in CTG signals, rather than model failure alone.

6.5. REAL WORLD CASE STUDIES AND DEPLOYMENT SCENARIOS

To demonstrate the practical applicability of the system, several real-world scenarios can be considered:

Case 1: Normal Pregnancy Monitoring

In cases where CTG data shows stable fetal heart rate, normal variability, and frequent accelerations, the model correctly predicts a Normal condition. SHAP explanations highlight positive contributions from these features, confirming a healthy fetal state. This allows clinicians to proceed with routine monitoring.

Case 2: Suspect Condition Detection

For cases with moderate deviations, such as reduced variability or occasional decelerations, the model classifies the condition as Suspect. SHAP analysis reveals mixed feature contributions, indicating uncertainty. This helps clinicians decide on closer observation or additional tests.

Case 3: Pathological Case Identification

In high-risk cases with significant abnormalities, such as prolonged decelerations and severely reduced variability, the model predicts a Pathological condition. SHAP explanations strongly emphasize these features, supporting the need for immediate medical intervention.

Discussion:

These case studies demonstrate that the system is capable of:

- Accurately identifying different levels of fetal risk
- Providing interpretable insights for each prediction
- Supporting timely and informed clinical decisions

The combination of predictive accuracy and explainability ensures that the system can be effectively used in real-world healthcare environments.

Chapter 7

Conclusion and Future Work

7.1 CONCLUSION

The proposed fetal health monitoring system successfully demonstrates the application of machine learning and explainable AI in the early detection of fetal distress using cardiotocography (CTG) data. The developed model achieves high accuracy and strong performance across multiple evaluation metrics, particularly in identifying critical Pathological cases, which is essential in clinical decision-making.

The integration of SHAP (SHapley Additive Explanations) is a significant contribution of this work, as it improves the interpretability and transparency of model predictions. The system bridges the divide between artificial intelligence and clinical practice by elucidating the impact of input features, such as fetal heart rate variability, accelerations, and decelerations, on the output. This interpretability is essential in the healthcare sector, where trust and accountability are paramount.

The system is intended to serve as a comprehensive decision-support tool, incorporating real-time prediction, bulk processing, and automated report generation, in addition to serving as a predictive model. These features enhance the system's usability and render it suitable for practical deployment in hospitals and clinics. Furthermore, the system employs techniques such as ensemble learning and SMOTE to resolve critical challenges, including class imbalance and model reliability.

Overall, the project is a promising solution for aiding healthcare professionals in the assessment of fetal health, as it accomplishes a balanced combination of interpretability, practical applicability, and accuracy. Although the model performs exceptionally well in Normal and Pathological cases, the Suspect class is inherently challenging due to overlapping characteristics, which reflect the complexity of real-world clinical situations..

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7.2 FUTURE WORK

Although the current system demonstrates strong performance, several enhancements can be explored to further improve its effectiveness and real-world applicability.

- **Integration with Real-Time IoT-Based Monitoring Systems**

  One of the major enhancements for the proposed system is the integration with real-time CTG monitoring devices using IoT technologies. This would enable continuous streaming of fetal health data directly into the system, allowing real-time analysis and instant alerts for critical conditions. Such an extension would significantly improve early detection capabilities and make the system more practical for use in intensive care and remote monitoring scenarios.

- **Incorporation of Advanced Deep Learning Models**

  While the current system uses a Random Forest classifier, future work can explore advanced deep learning approaches such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Transformer-based models. These models are particularly effective in capturing temporal dependencies in sequential CTG data, which can further enhance prediction accuracy and improve detection of subtle patterns in fetal heart rate signals.

- **Improvement in Suspect Class Classification**

  The Suspect class remains challenging due to overlapping characteristics with Normal and Pathological classes. Future improvements can focus on using hybrid models, ensemble stacking techniques, or probabilistic classification methods to better handle borderline cases. Additionally, threshold tuning and confidence-based prediction mechanisms can be incorporated to reduce ambiguity in classification.

- **Expansion of Dataset and Clinical Validation**

  The performance of the system can be further improved by training on larger and more diverse datasets collected from real-world clinical environments. Including multi-center hospital data will enhance the model's generalization ability. Furthermore, clinical validation through collaboration with healthcare professionals can ensure the system's reliability and acceptance in real-world applications.

- **Development of Interactive Explainability Dashboards**

  Although SHAP provides strong interpretability, future work can focus on building interactive dashboards where clinicians can explore both global and local explanations in real time. Features such as patient-specific SHAP plots, feature comparison tools, and dynamic visualization will improve usability and make the system more intuitive for medical practitioners.

- **Integration with Multi-Modal Healthcare Data**

  The current system relies solely on CTG data. Future extensions can include integration with additional data sources such as ultrasound imaging, maternal health records, and laboratory reports. Combining multiple data modalities can provide a more comprehensive understanding of fetal health and improve overall diagnostic accuracy.

- **Deployment as a Scalable Cloud-Based Application**

  To increase accessibility, the system can be deployed as a cloud-based platform, enabling remote access for hospitals, clinics, and telemedicine services. This would allow multiple users to access the system simultaneously and support large-scale data processing, making it suitable for widespread adoption.

- **Enhancement of Automated Report Generation**

  Future improvements can include more detailed and customizable report generation, incorporating clinical recommendations, risk scoring, and longitudinal analysis of patient data. This would further assist clinicians in making informed decisions and improve documentation quality.

- **Ensuring Data Privacy and Regulatory Compliance**

  For real-world deployment, it is essential to ensure compliance with healthcare data protection regulations. Future work should focus on implementing secure data handling mechanisms, encryption techniques, and adherence to medical standards to ensure patient data privacy and system reliability.

Appendices

Appendix 1
Dataset Information

The dataset used in this study is sourced from kaggle:
https://www.kaggle.com/datasets/andrewmvd/fetal-health-classification

1. Dataset Overview

- The dataset contains **cardiotocography (CTG) recordings** used to assess fetal health.
- It includes **2126 instances** with **21 input features** and **1 target variable**.
- Each record represents features extracted from fetal heart rate signals and uterine contractions.
- The dataset is widely used in medical ML research for classification of fetal conditions.

2. Target Variable

- **fetal_health (Output Class):**
  - 1 → Normal
  - 2 → Suspect
  - 3 → Pathological

3. Feature Description (Important Features)

- Baseline Value → Average fetal heart rate (FHR)
- Accelerations → Increase in FHR (healthy sign)
- Fetal Movement → Movement detected from fetus
- Uterine Contractions → Contraction activity
- Decelerations (light/severe/prolonged) → Drops in FHR (risk indicators)
- Short-Term Variability (STV) → Beat-to-beat variability
- Long-Term Variability (LTV) → Overall variability trend
- Histogram Features → Statistical representation of FHR distribution

Appendix 2
Data Preprocessing Steps

Data preprocessing is a critical step in ensuring the quality and reliability of the machine learning model. The following steps were systematically applied to prepare the dataset for training and evaluation.

1. Data Inspection and Understanding

- The dataset was first examined to understand:
  - Feature types (numerical variables)
  - Target variable distribution (Normal, Suspect, Pathological)
- Summary statistics such as mean, standard deviation, minimum, and maximum values were analyzed.
- This step helped identify:
  - Scale differences between features
  - Potential anomalies in CTG measurements

2. Handling Missing Values

- The dataset was checked for missing or null values.
- Observation:
  - No missing values were present in the dataset.
- Therefore:
  - No imputation techniques (mean/median filling) were required.
- This ensured data integrity and avoided bias due to artificial value replacement.

3. Feature Scaling (Normalization / Standardization)

- Since features such as:
  - Baseline value (range ~120–160)
  - Accelerations (very small decimal values)
  - Histogram features (large numeric ranges) have different scales, scaling was necessary.

- **Standardization (Z-score normalization)** was applied:

$$Z = \frac{X - \mu}{\sigma}$$

Where:

- **X** = original feature value
- **μ** = mean of the feature
- **σ** = standard deviation

Purpose:

- Brings all features to a common scale
- Prevents dominance of large-value features
- Improves model performance and convergence

4. Feature-Target Separation

- The dataset was divided into:
  - Input features (X) → All CTG parameters
  - Target variable **(y)** → Fetal health class

Representation:

- $X = \{x1,x2,...,xn\}$
- $y \in \{1,2,3\}$

Where:

- 1 → Normal
- 2 → Suspect
- 3 → Pathological

5. Train-Test Splitting

- The dataset was divided into:
  - Training set (80%)
  - Testing set (20%)

Purpose:

- Training set → Used to train the model
- Testing set → Used to evaluate performance on unseen data

Advantage:

- Prevents overfitting
- Ensures generalization capability

6. Handling Class Imbalance

- The dataset contains:
  - More Normal cases
  - Fewer Suspect and Pathological cases

Issue:

- Model may become biased toward the majority class

Approach Used:

- Stratified splitting was applied to maintain class proportions in both training and testing datasets.

7. Feature Importance Awareness (Pre-Model Insight)

- Certain features were identified as clinically significant:
  - Prolonged decelerations
  - Abnormal short-term variability
  - Percentage of abnormal long-term variability

Purpose:

- Helps in:
  - Understanding model behavior
  - Supporting explainability analysis (SHAP)

8. Data Consistency and Validation

- Ensured:
  - No duplicate records
  - All values fall within medically valid ranges
- Checked for:
  - Logical consistency (e.g., no negative physiological values)

9. Final Preprocessed Dataset

- Output dataset characteristics:
  - Clean (no missing values)
  - Scaled features
  - Balanced class representation (via stratification)
  - Ready for model training

Appendix 3
Model Implementation Details

The model implementation focuses on designing a robust and interpretable classification system for fetal health assessment using cardiotocography (CTG) features. A machine learning pipeline was developed to ensure efficient data handling, accurate prediction, and reliable generalization.

MODEL ARCHITECTURE

The proposed system follows a structured pipeline consisting of data input, preprocessing, model training, and prediction stages.

1. Input Layer (Feature Set)

- Accepts 21 CTG features extracted from fetal monitoring data
- Includes:
  - Fetal heart rate parameters (baseline, accelerations, decelerations)
  - Variability measures (short-term, long-term)
  - Histogram-based statistical features

2. Preprocessing Layer

- Applies:
  - **Standardization (Z-score normalization)** to ensure uniform feature scaling
- Maintains:
  - Consistent data distribution across training and testing sets
- Ensures:
  - Improved learning efficiency and model stability

3. Data Balancing Layer

- Uses **SMOTE (Synthetic Minority Oversampling Technique)**
- Purpose:
  - Address class imbalance between Normal, Suspect, and Pathological cases
- Outcome:
  - Enhanced learning of minority class patterns

4. Core Model Layer (Random Forest Classifier)

- Consists of multiple decision trees trained in parallel
- Each tree learns from a random subset of data and features
- Final prediction: Determined through majority voting mechanism

Key Characteristics:

- Ensemble-based learning
- Handles non-linear relationships
- Reduces overfitting through averaging

5. Prediction Layer

- Outputs:
  - **Class label** (Normal / Suspect / Pathological)
  - **Probability scores** for each class

$$\hat{y} = \arg\max P(y = k|x)$$

- Enables:
  - Risk-level interpretation (Low / Moderate / High)

6. Evaluation Layer

- Model performance assessed using:
  - Accuracy
  - Precision
  - Recall
  - F1-score
  - ROC-AUC
- Ensures:
  - Reliable classification across all classes

Model Workflow Summary

- Input CTG data → Preprocessing → Data balancing → Model training → Prediction → Evaluation

TRAINING CONFIGURATION

Parameter	Value / Setting	Description
Model Type	Random Forest Classifier	Ensemble-based supervised learning algorithm
Number of Trees (n_estimators)	200	Total decision trees used in the forest for better stability
Maximum Depth (max_depth)	8	Controls tree depth to prevent overfitting
Splitting Criterion	Gini Impurity	Measures quality of splits during tree construction
Random State	42	Ensures reproducibility of results
Feature Selection	Random Subset per Split	Introduces randomness to improve generalization
Data Scaling	Standardization (Z- score)	Normalizes feature values to common scale
Train-Test Split	80% Train / 20% Test	Used for model training and evaluation
Sampling Strategy	SMOTE Applied	Balances minority classes (Suspect & Pathological)
Validation Strategy	Hold-out Validation	Evaluates model on unseen test data
Input Features	21 CTG Features	Includes FHR, variability, histogram metrics
Output Classes	3 Classes	Normal, Suspect, Pathological
Evaluation Metrics	Accuracy, Precision, Recall, F1-score, ROC-AUC	Measures classification performance
Probability Output	Enabled	Model predicts class probabilities
Overfitting Control	Depth Limitation + Ensemble Averaging	Reduces variance and improves robustness

Appendix 4

Explainable AI (SHAP) Integration

The integration of Explainable AI is essential to enhance the transparency and clinical reliability of the proposed fetal health prediction system. SHAP (SHapley Additive exPlanations) is incorporated to interpret model decisions by quantifying the contribution of each feature to the final prediction.

PROCESS WORKFLOW:

1. **Model Input Processing**: Preprocessed CTG feature values are passed to the trained model.
2. **SHAP Explainer Initialization**: A Tree-based SHAP explainer is used, compatible with the Random Forest model.
3. **SHAP Value Computation**: SHAP values are calculated for each feature, representing their contribution to the prediction.
4. **Local Explanation Generation**: For an individual patient, SHAP identifies:
   - Features pushing prediction toward Normal
   - Features pushing toward Pathological
5. **Global Interpretation**: Aggregated SHAP values across the dataset highlight:
   - Most influential features
   - Overall model behavior trends

OUTPUT STORAGE:

SHAP values are:

- Pre-computed for validation samples to reduce runtime overhead
- Stored along with: Input feature values, Model predictions

This enables: Faster visualization, Reusable explainability insights, Integration into dashboards and reports

SIGNIFICANCE:

- **Clinical Interpretability**: Helps healthcare professionals understand *why* a prediction was made
- **Trust and Transparency**: Reduces “black-box” nature of machine learning models
- **Decision Support**: Highlights critical risk factors such as:
  - Prolonged decelerations
  - Abnormal variability

Appendix 5
Software and Hardware Environment

The successful implementation and evaluation of the proposed fetal health prediction system required a well-defined software and hardware environment. The selected tools and configurations ensure efficient data processing, model training, visualization, and deployment.

SOFTWARE STACK

Component	Description
Programming Language	Python 3.x
Machine Learning Framework	Scikit-learn
Explainability Tool	SHAP (TreeExplainer)
Data Processing Libraries	NumPy, Pandas
Visualization Tools	Matplotlib, Seaborn, Plotly
Web Framework	Streamlit
Model Persistence	Joblib
Report Generation	ReportLab
Development Environment	Visual Studio Code
Operating System	Windows

HARDWARE CONFIGURATION

Specification	Details
Processor	Intel Core i5 or higher
RAM	8 GB (Recommended)
Storage	256 GB SSD or higher
GPU	Not required (CPU sufficient)
System Type	64-bit architecture

- The system was developed using Visual Studio Code as the primary development environment, enabling efficient coding, debugging, and integration.
- Model training, evaluation, and testing were performed locally using Python- based libraries.
- The application interface was built using Streamlit, allowing real-time interaction and visualization.

Appendix 6
Ethical Considerations and Limitations

The project adheres strictly to ethical research and data privacy standards. The dataset was publicly available through Kaggle and contains no patient-identifiable information. It was used solely for academic and non-commercial purposes.

ETHICAL ASPECTS:

- Patient Data Privacy: Data must be anonymized before use
- Clinical Responsibility: AI should assist, not replace doctors
- Bias in Dataset: Dataset may not represent all populations
- Explainability Requirement: Models must be interpretable in healthcare

LIMITATIONS:

- Model trained on secondary dataset (not real-time data)
- Limited ability to capture temporal patterns
- Suspect class classification remains difficult
- No integration with live CTG devices
- Clinical validation not performed

FUTURE ETHICAL WORK:

Future efforts should include cross-institutional validation, integration into clinical decision systems with obsetrician oversight, and ensuring model fairness across patient demographics.

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